



ARDUINO IoT KIT

• CODE • CONNECT • INNOVATE •



Explore
connected
devices



Work with
sensor data



Build smart
projects



Arduino IOT Kit

What is an Arduino IOT KIT ?

An Arduino IoT Kit is a collection of electronic components designed to help users build smart devices that can connect to the internet and interact with the real world. It typically includes a microcontroller board such as Arduino along with various sensors, actuators, and communication modules like Wi-Fi or Bluetooth. These components allow the system to collect data and send it to online platforms where users can monitor and control their projects remotely. Arduino IoT Kits are widely used in applications like smart home automation, environmental monitoring, and industrial systems. They are especially useful for beginners and students because they provide a simple and hands-on way to learn electronics, programming, and Internet of Things (IoT) concepts.

Why do we need an Arduino IoT Kit?

1.To Learn Core Concepts: Students understand important ideas such as:

How sensors collect data

How devices communicate over the internet

Basics of microcontrollers and connectivity

Hands-on Learning Experience: Instead of only theory, students build real IoT projects, experiment with different components, and gain practical and technical skills.

To Build Smart Applications: Using the kit, students can create home automation systems, weather monitoring devices, and remote-controlled gadgets.

Enhances Problem-Solving Skills: While working on projects, students debug errors in code and fix circuits, think logically step-by-step, and improve critical and analytical thinking.

Kit Components Overview

Arduino Uno R3: The Arduino Uno R3 is a widely used microcontroller board based on the ATmega328P. It serves as the main control unit that processes inputs from sensors and sends outputs to devices like LEDs and motors. It can be easily programmed using the Arduino IDE with simple C/C++ code. This board is ideal for beginners as well as advanced embedded system projects.

USB Cable: A USB cable is used to connect the Arduino board to a computer for programming and power. It enables uploading of code from the Arduino IDE to the microcontroller. Additionally, it supplies power to the board during testing and operation. It is an essential accessory for communication and development.

Jumper Wires: Jumper wires are used to connect components on a breadboard or Arduino. They come in different types like male-to-male, male-to-female, and female-to-female. These wires help in creating temporary electrical connections. They are essential for organizing and building circuits neatly.

Resistors Kit: Resistors are components that limit the flow of current in a circuit. They are used to protect sensitive components like LEDs from damage. A resistor kit contains different resistance values for various applications. They are fundamental in controlling voltage and current in circuits.

LEDs (various colors): LEDs are light-emitting diodes that glow when current passes through them. They are commonly used as indicators or for decorative lighting. Different colors represent different voltage requirements. LEDs require a resistor to avoid excess current and damage.

RGB LED: An RGB LED contains red, green, and blue LEDs in one package. By varying the intensity of each color, it can produce a wide range of colors. It is widely used in displays and lighting effects. The Arduino controls it using PWM signals.

Push Buttons: Push buttons are simple input devices used to interact with a circuit. When pressed, they complete or break a circuit connection. They are commonly used to send commands to the Arduino. These are widely used in user-controlled electronic projects.

Buzzer (Active/Passive): A buzzer is an electronic device that produces sound. Active buzzers generate sound when powered, while passive buzzers require a signal. They are used in alarms, notifications, and alert systems. They are easy to interface with Arduino.

HC-05 Bluetooth Module: The HC-05 Bluetooth module allows wireless communication between Arduino and devices like smartphones. It uses serial communication through TX and RX pins. It is commonly used in remote control and automation systems. This module simplifies wireless data transfer in projects.

ESP8266 ESP-12E: The ESP8266 ESP-12E is a Wi-Fi module used to connect Arduino projects to the internet. It supports TCP/IP networking and IoT applications. It can also function as a standalone microcontroller. It is widely used in smart home and cloud-based projects.

Ultrasonic Sensor (HC-SR04): The ultrasonic sensor measures distance using sound waves. It sends ultrasonic pulses and calculates distance based on echo time. It is commonly used in obstacle detection and robotics. It provides accurate and reliable measurements.

Temperature Sensor (DHT11): The DHT11 sensor measures temperature and humidity in the environment. It provides digital output, making it easy to read data. It is commonly used in weather monitoring systems. It is affordable and simple to use.

LDR (Light Sensor): An LDR detects light intensity by changing its resistance. The resistance decreases as light increases. It is used in automatic lighting and brightness control systems. It works with analog signals in Arduino.

IR Sensor Module: The IR sensor detects infrared light or nearby objects. It is commonly used in line-following robots and obstacle detection. It provides digital output signals. It is simple and efficient for sensing applications.

PIR Motion Sensor: The PIR sensor detects motion by sensing infrared radiation from humans. It is widely used in security and automation systems. It triggers when movement is detected. It is energy-efficient and reliable.

Soil Moisture Sensor: This sensor measures the moisture level in soil. It is used in automatic plant watering systems. It helps in maintaining proper soil conditions. It provides analog or digital output.

Gas Sensor: A gas sensor detects harmful gases like LPG or smoke. It is used for safety and monitoring purposes. It provides output based on gas concentration. It is commonly used in gas leakage detection systems.

Rain Drop Sensor: The rain sensor detects water droplets on its surface. It works based on conductivity changes. It is used in weather monitoring and automation systems. It triggers when rain is detected.

Touch Sensor: A touch sensor detects human touch using capacitance. It replaces traditional mechanical buttons. It provides a digital output signal. It is used in modern user interfaces.

Water Pump: A water pump is used to move water in projects. It is commonly used in irrigation and water circulation systems. It is controlled using relays or motor drivers. It operates on DC power.

Infrared IR Wireless Remote Control Module: An IR wireless remote control module is used to send and receive signals without wires using infrared light. It is commonly used in TVs, air conditioners, and DIY electronics projects to control devices remotely. It consists of an IR transmitter (remote) and an IR receiver module. It is controlled using a microcontroller like Arduino by decoding the received signals. It operates on low-voltage DC power.

Relay Module: A relay module acts as an electrically controlled switch. It allows Arduino to control high-voltage devices safely. It provides isolation between circuits. It is widely used in automation systems.

Servo Motor :The servo motor provides precise angle control. It can rotate typically between 0° and 180°. It is used in robotics and automation. It is controlled using PWM signals.

DC Motor: A DC motor converts electrical energy into mechanical motion. It is used in fans, wheels, and machines. Its speed can be controlled using PWM. It requires a motor driver for proper control.

Fan Module: A fan module is used for cooling or airflow in projects. It works similarly to a DC motor with attached blades. It can be controlled using Arduino. It is used in temperature-based systems.

Bo Motor: A BO motor is a geared DC motor used in robots. It provides high torque and controlled speed. It is ideal for robot car movement. It works well with motor drivers.

Motor Driver (L293D):The L293D controls motors using Arduino signals. It allows forward and reverse motion. It protects Arduino from high current. It is essential for motor-based projects.

Wheels: Wheels are attached to motors for movement. They help robots move on different surfaces. They provide stability and smooth motion. They are essential for mobile robots.

Caster Wheel: A caster wheel supports the robot structure. It rotates freely in any direction. It helps maintain balance during movement. It is used in robot cars.

16x2 LCD Display with I2C: This display shows text in 2 lines with 16 characters each. The I2C module reduces the number of wires needed. It is commonly used for displaying sensor data. It is easy to interface with Arduino.

TM1637 4 Digit 7 Segment Display: This display shows numeric values using four digits. It uses the TM1637 driver for easy control. It requires only two wires for communication. It is commonly used in clocks and counters.

9 V Battery

A 9V battery provides portable power to circuits. It is useful when external power is not available. It is easy to connect and use. However, it has limited current capacity.

Battery Clip

A battery clip connects a 9V battery to a circuit. It ensures a secure and reliable connection. It usually has two wires for positive and negative terminals. It is easy to use and replace.

DC Jack Clip

A DC jack clip allows connection of a battery to the Arduino power jack. It ensures proper polarity and stable power supply. It is commonly used with 9V batteries. It provides a convenient power interface.

Safety Instructions

1. Always check all circuit connections before powering the system.
2. Use a proper and compatible power supply for the Arduino board.
3. Avoid short circuits by placing wires correctly on the breadboard.
4. Maintain correct polarity while connecting LEDs, sensors, and batteries.
5. Do not exceed voltage and current ratings of components.
6. Disconnect power before making any changes to the circuit.
7. Do not touch exposed wires or components while the circuit is powered..
8. Turn off power immediately if you notice overheating, smoke, or unusual behavior.

Projects

1. Bluetooth Smart Home Control

Components Required

- ❖ Arduino Uno
- ❖ HC-05 / HC-06 Bluetooth Module
- ❖ LED (for light demo)
- ❖ Relay Module (for fan/light control)
- ❖ Buzzer
- ❖ 220Q Resistor (for LED)
- ❖ Jumper wires, Breadboard

Connections

Bluetooth Module (HC-05)

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ TX → Arduino Pin 10 (RX)
- ❖ RX → Arduino Pin 11 (TX) (*use voltage divider if needed*) **LED (Light)**
- ❖ Pin 13 → 2200 Resistor ^ LED Anode
- ❖ LED Cathode → GND

Relay (Fan / Bulb)

- ❖ IN → Pin 8
- ❖ VCC → 5V
- ❖ GND ^ → GND

Buzzer

- ❖ Positive → Pin 9
- ❖ Negative → GND

Code Example

Scan the QR

Mobile App Control

You can use apps like:

- Bluetooth Terminal
- Arduino Bluetooth Controller

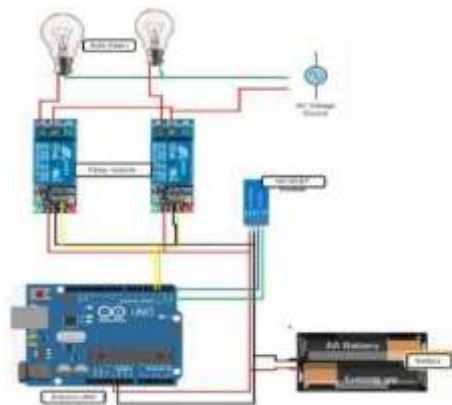
Buttons in app send characters:

- | | |
|-----|-------------|
| • A | ^ Light ON |
| • a | Light OFF |
| • B | ^ Fan ON |
| • b | ^ Fan OFF |
| • C | ^ Buzzer ON |
| • c | Buzzer OFF |

Working Explanation Step-by-Step Flow

1. Mobile app sends command via Bluetooth
2. HC-05 receives signal
3. Arduino reads the character
4. Based on command:
 - o Turns ON/OFF LED
 - o Activates relay (fan/light)
 - o Turns buzzer ON/OFF

Reference Image



2. Bluetooth Car

Components Required

- ❖ Arduino Uno
- ❖ HC-05 / HC-06 Bluetooth Module
- ❖ L298N Motor Driver Module
- ❖ 2 DC Motors + Wheels
- ❖ Robot Chassis
- ❖ Battery (7.4V / 9V)
- ❖ Jumper wire

Connections

Bluetooth Module (HC-05)

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ TX → Pin 10 (Arduino RX)
- ❖ RX → Pin 11 (Arduino TX)

Motor Driver (L298N)

Control Pins

- ❖ IN1 → Pin 2
- ❖ IN2 → Pin 3
- ❖ IN3 → Pin 4
- ❖ IN4 → Pin 5

Enable Pins

- ❖ ENA → Pin 6 (PWM)
- ❖ ENB → Pin 9 (PWM)

Power

- ❖ 12V → Battery +
- ❖ GND → Battery - and Arduino GND **Motors**
- ❖ Left Motor → OUT1 & OUT2
- ❖ Right Motor → OUT3 & OUT4

Mobile App Control

You can use joystick-based apps like:

- Arduino Bluetooth Controller
- Bluetooth RC Controller

Joystick Commands

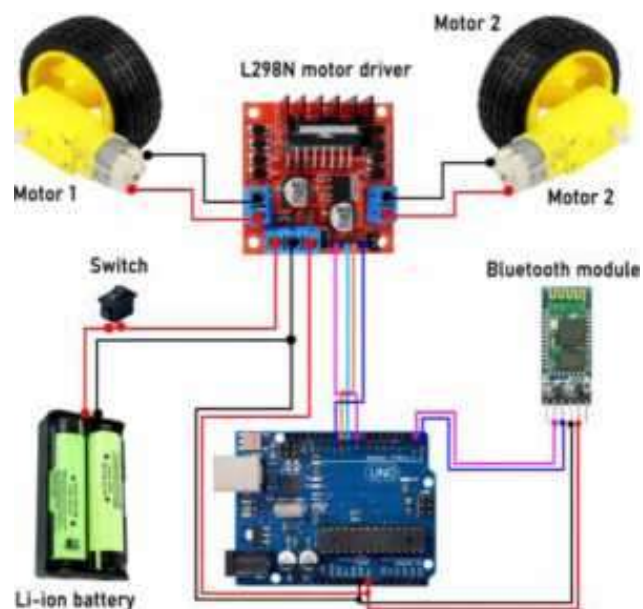
- Forward → 'F'
- Backward → 'B'
- Left → 'L'
- Right → 'R'
- Stop → 'S'

Working Explanation

Step-by-Step Flow

1. You move joystick on mobile app
2. App sends direction command via Bluetooth
3. HC-05 receives signal
4. Arduino reads command
5. Motor driver controls motors
6. Car moves accordingly

Reference Image



3. Bluetooth Password Door Lock

Components Required

- ❖ Arduino Uno
- ❖ HC-05 / HC-06 Bluetooth Module
- ❖ Servo Motor (SG90 or similar)
- ❖ Buzzer (optional for alert)
- ❖ Jumper wires
- ❖ Breadboard

Connections

Bluetooth Module (HC-05)

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ TX → Pin 10 (Arduino RX)
- ❖ RX → Pin 11 (Arduino TX)

Servo Motor (Door Lock)

- ❖ Red → 5V
- ❖ Brown/Black → GND
- ❖ Yellow/Orange → Pin 9

Buzzer (Optional)

- ❖ Positive → Pin 8
- ❖ Negative → GND

You can use apps like:

- ❖ Bluetooth Terminal
- ❖ Serial Bluetooth Terminal

How to Send Password

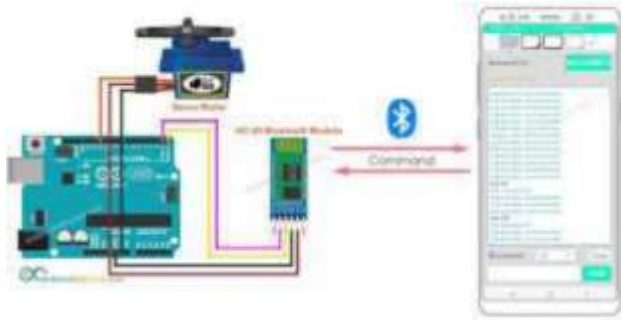
Type password and send with # at end:

Example: 1234#

Working Explanation Step-by-Step Flow

1. User enters password in mobile app
2. Bluetooth sends data to Arduino
3. Arduino stores characters
4. When # is received ^ check password
5. If correct:
 - o Servo rotates ^ door unlocks
 - o Waits ^ locks again
6. If wrong:
 - o Buzzer sounds

Reference Image



4. Bluetooth Voice-Controlled System

Components Required

- ❖ Arduino Uno
- ❖ HC-05 / HC-06 Bluetooth Module
- ❖ Relay Module (for appliances like fan/light)
- ❖ LED (for demo)
- ❖ Buzzer (optional)
- ❖ Jumper wires
- ❖ Breadboard

Connections

Bluetooth Module (HC-05)

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ TX → Pin 10 (Arduino RX)
- ❖ RX → Pin 11 (Arduino TX)

Relay Module (Appliances)

- ❖ IN1 → Pin 7 (Light)
- ❖ IN2 → Pin 8 (Fan)
- ❖ VCC → 5V
- ❖ GND → GND

LED (Optional Demo)

❖ Pin 13 → 220ohm → LED → GND

Buzzer (Optional)

❖ Positive → Pin 9

❖ Negative → GND

Mobile App Control

Use voice-control apps like:

❖ Arduino Voice Control

❖ BT Voice Control for Arduino

Voice Commands Examples

Voice Command	Action
"light on"	Turn ON light
"light off"	Turn OFF light
"fan on"	Turn ON fan
"fan off"	Turn OFF fan
"buzzer on"	Activate buzzer
"buzzer off"	Stop buzzer

Working Explanation

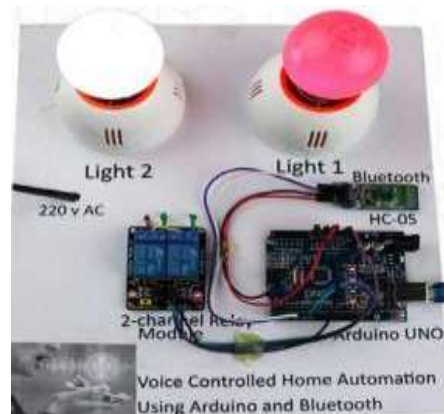
Step-by-Step Flow

1. User speaks command into mobile app
2. App converts voice ^ text
3. Text sent via Bluetooth
4. HC-05 receives data
5. Arduino reads full string
6. Matches command
7. Executes corresponding action

System Flow

Voice → Mobile App → Bluetooth → Arduino → Relay → Appliance

Reference Image



5. Bluetooth RGB LED Controller

Components Required

- ❖ Arduino Uno
- ❖ HC-05 / HC-06 Bluetooth Module
- ❖ RGB LED (Common Cathode recommended)
- ❖ 3 x 220Q Resistors
- ❖ Jumper wires
- ❖ Breadboard

Connections

Bluetooth Module (HC-05)

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ TX → Pin 10 (Arduino RX)
- ❖ RX → Pin 11 (Arduino TX)

RGB LED Connections

(Common Cathode type)

- ❖ Red Pin → Pin 9 (PWM) via 2200
- ❖ Green Pin → Pin 10 (PWM) via 2200 *(or use another PWM pin like 6 if conflict)*
- ❖ Blue Pin → Pin 11 (PWM) via 2200
- ❖ Common Cathode → GND

Mobile App Control

Use slider-based apps like:

- ❖ Arduino Bluetooth Controller
- ❖ RGB LED Controller

Slider Input Format

The app sends values like:

- ❖ 255 0 0 → Red
- ❖ 0 255 0 → Green
- ❖ 0 0 255 → Blue
- ❖ 255 255 0 → Yellow

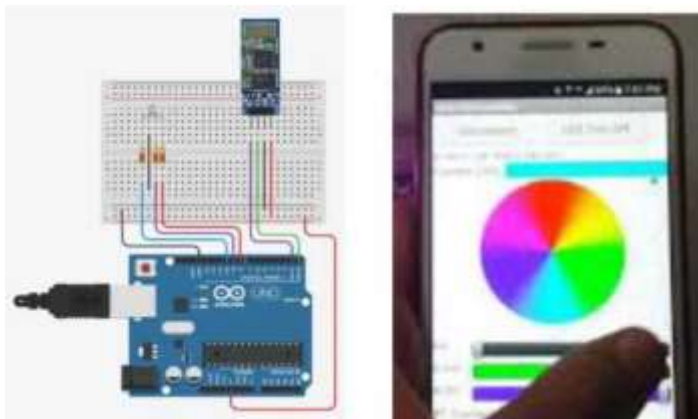
Working Explanation Step-by-Step Flow

- ❖ User adjusts RGB sliders on mobile
- ❖ App sends 3 values (R, G, B)
- ❖ Bluetooth module receives data
- ❖ Arduino reads values using `parseInt()`
- ❖ PWM signals generated using `analogWrite()`
- ❖ RGB LED glows in selected color

Color Mixing Logic

R	G	B	Color
255	0	0	Red
0	255	0	Green
0	0	255	Blue
255	255	0	Yellow
255	255	255	White
0	0	0	OFF

Reference Image



6. IoT Weather Monitoring System

Components Required

- ❖ Arduino Uno
- ❖ ESP-12E WiFi Module
- ❖ DHT11 Temperature & Humidity Sensor
- ❖ 10kΩ Resistor (for DHT)
- ❖ Breadboard
- ❖ Jumper wires

Connections

ESP8266 (ESP-12E)

Important: ESP8266 works at **3.3V only**

- ❖ VCC → 3.3V
- ❖ GND → GND
- ❖ TX → Arduino RX (Pin 0)
- ❖ RX → Arduino TX (Pin 1) (*via voltage divider*)
- ❖ CH_PD (EN) → 3.3V

Web Dashboard Platforms

You can send data to:

- ❖ ThingSpeak
- ❖ Blynk
- ❖ Adafruit IO

What You See on Dashboard

- ❖ Temperature graph
- ❖ Humidity graph
- ❖ Real-time updates
- ❖ Historical data trends

Working Explanation Step-by-Step Flow

1. DHT sensor reads temperature & humidity
2. Arduino processes the data
3. ESP8266 connects to WiFi
4. Data sent to cloud server (e.g., ThingSpeak)
5. Dashboard updates in real time

System Flow

Sensor → Arduino → ESP8266 → Internet → Cloud → Web Dashboard

7. IoT Smart Irrigation System

Components Required

- ❖ Arduino Uno
- ❖ ESP8266 WiFi Module (ESP-12E)
- ❖ Soil Moisture Sensor
- ❖ Relay Module
- ❖ Water Pump / Motor
- ❖ Jumper wires
- ❖ Breadboard
- ❖ Power supply (for pump)

Connections

Soil Moisture Sensor

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ Analog Output → A0

Relay Module (Pump Control)

- ❖ IN → Pin 7
- ❖ VCC → 5V
- ❖ GND → GND

Water Pump

- ❖ Connected through relay (external power supply)

ESP8266 (WiFi)

- ❖ VCC → 3.3V
- ❖ GND → GND
- ❖ TX → Arduino RX
- ❖ RX → Arduino TX

IoT Platforms (Remote Control & Monitoring)

You can connect this system to:

- ❖ Blynk
- ❖ ThingSpeak
- ❖ Adafruit IO

Remote Features

- ❖ View soil moisture data
- ❖ Turn pump ON/OFF manually
- ❖ Set threshold values
- ❖ Receive alerts

Working Explanation Step-by-Step Flow

1. Soil moisture sensor reads soil condition
2. Arduino processes analog value
3. If soil is dry:
 - o Relay activates o Pump starts watering
4. If soil is wet:
 - o Pump stops
5. Data sent to cloud via WiFi
6. User monitors/controls from mobile

System Flow

Soil Sensor → Arduino → Relay → ESP8266 → Internet → Mobile Dashboard

8. IoT Home Automation

Components Required

- ❖ Arduino UNO
- ❖ ESP8266 WiFi Module (ESP-12E)
- ❖ Relay Module (2-channel)
- ❖ LED / Bulb / Fan
- ❖ Jumper wires
- ❖ Breadboard
- ❖ External power supply (for relay load)

Connections

ESP8266 → Arduino UNO

- ❖ ESP TX → Arduino RX (Pin 0)
- ❖ ESP RX → Arduino TX (pin 1)
- ❖ VCC → 3.3V
- ❖ GND → GND
- ❖ EN (CH_PD) → 3.3V

Relay Module

- ❖ IN1 → Arduino Pin 7 → Light
- ❖ IN2 → Arduino Pin 8 → Fan
- ❖ VCC → 5V
- ❖ GND → GND

Power

- ❖ Arduino → USB
- ❖ ESP8266 → Stable 3.3V
- ❖ Appliances → External AC via relay

How to Use

1. Upload Arduino code to UNO
2. Upload WiFi code to ESP8266
3. Power both modules
4. Connect ESP8266 to WiFi
5. Open Serial Monitor ^ get IP address
6. Enter IP in browser (e.g., 192.168.x.x)
7. Click buttons ^ control devices

What the Web Page Does

1. Light ON/OFF
2. Fan ON/OFF
3. Can be accessed from phone/laptop browser **Working Explanation**

Step-by-Step Flow

1. ESP8266 connects to WiFi
2. Starts web server
3. User opens IP in browser
4. Webpage appears with buttons
5. When button clicked:
 - o HTTP request sent
6. ESP8266 reads request
7. Sends command via Serial to Arduino
8. Arduino controls relay
9. Appliance turns ON/OFF instantly

System Flow

User Browser → WiFi → ESP8266 → Serial → Arduino UNO → Relay → Appliance

URL-Based Control

URL	Action
lightON	Light ON
lightOFF	Light OFF
fanON	Fan ON
fanOFF	Fan OFF

9. IoT Gas Leakage Alert System

Components Required

- ❖ Arduino UNO
- ❖ ESP8266 WiFi Module (ESP-12E / ESP-01)
- ❖ MQ-2 Gas Sensor
- ❖ Buzzer
- ❖ LED (optional)
- ❖ Jumper wires
- ❖ Breadboard

Connections MQ-2 Gas Sensor

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ AO → A0 (Arduino UNO)

Buzzer

- ❖ Positive → Pin 8 (Arduino)
- ❖ Negative → GND

LED (Optional)

- ❖ Anode → Pin 9 (via 2200 resistor)
- ❖ Cathode → GND

ESP8266 → Arduino UNO

- ❖ ESP TX → Arduino RX (Pin 0)
- ❖ ESP RX → Arduino TX (pin 1)
- ❖ VCC → 3.3V
- ❖ GND → GND
- ❖ EN (CH_PD) → 3.3V

Notification Platforms

You can send alerts using:

- ❖ Blynk
- ❖ Telegram
- ❖ IFTTT

What Happens During Alert

- ❖ Buzzer sounds locally
- ❖ LED turns ON
- ❖ Notification sent to phone
- ❖ Warning message displayed
- ❖ Anode → Pin 9 (via 2200 resistor)
- ❖ Cathode → GND

ESP8266 → Arduino UNO

- ❖ ESP TX → Arduino RX (Pin 0)
- ❖ ESP RX → Arduino TX (pin 1)
- ❖ VCC → 3.3V
- ❖ GND → GND
- ❖ EN (CH_PD) → 3.3V

Notification Platforms

You can send alerts using:

- ❖ Blynk
- ❖ Telegram
- ❖ IFTTT

What Happens During Alert

- ❖ Buzzer sounds locally
- ❖ LED turns ON
- ❖ Notification sent to phone
- ❖ Warning message displayed

Working Explanation Step-by-Step Flow

- ❖ MQ-2 sensor detects gas concentration
- ❖ Arduino UNO reads analog value
- ❖ If gas level exceeds threshold:
 - o Buzzer turns ON
 - o LED turns ON
 - o Sends "ALERT" to ESP8266
- ❖ ESP8266 receives data via Serial
- ❖ Connects to internet
- ❖ Sends notification (IFTTT/Blynk/Telegram)
- ❖ User receives alert on mobile

System Flow

Gas Sensor → Arduino UNO → Serial → ESP8266 → WiFi → Cloud → Mobile

Detection Logic

Gas Level	Sensor Value	Action
Safe	Low (<400)	No alert
Dangerous	High (>400)	Alert + Buzzer

10.IoT Energy Monitoring System

Components Required

- ❖ Arduino UNO
- ❖ ESP8266 WiFi Module (ESP-12E / ESP-01)
- ❖ Potentiometer (*simulation*)
- ❖ Jumper wires
- ❖ Breadboard

Connections

Potentiometer (Simulation)

- ❖ One end →5V
- ❖ Other end →GND
- ❖ Middle (wiper) →A0 (Arduino UNO)

ESP8266 → Arduino UNO

- ❖ ESP TX →Arduino RX (Pin 0)
- ❖ ESP RX → Arduino TX (pin 1)
- ❖ VCC →3.3V
- ❖ GND →GND
- ❖ EN (CH_PD) → 3.3V

Power

- ❖ Arduino →USB
- ❖ ESP8266 →Stable 3.3V

Cloud Platforms

You can visualize data using:

- ❖ ThingSpeak
- ❖ Blynk
- ❖ Adafruit IO

Dashboard Output

- ❖ Power consumption graph
- ❖ Real-time updates
- ❖ Historical data tracking

Working Explanation Step-by-Step Flow

1. Potentiometer simulates energy input
2. Arduino UNO reads analog value
3. Converts value → voltage → power
4. Sends data via Serial to ESP8266
5. ESP8266 connects to WiFi
6. Uploads data to cloud platform
7. Dashboard updates with new readings

System Flow

Sensor → Arduino UNO → Serial → ESP8266 → WiFi → Cloud → Dashboard

Calculation Logic

Step	Formula
Analog Read	0-1023
Voltage	$(\text{value} / 1023) \times 5$
Power	Voltage x Constant

11.IoT Rain Monitoring Station

Components Required

- ❖ Arduino UNO
- ❖ ESP8266 WiFi Module (ESP-12E / ESP-01)
- ❖ Rain Sensor Module (YL-83 / FC-37)
- ❖ Jumper wires
- ❖ Breadboard

Connections

Rain Sensor Module

- VCC → 5V
- GND → GND
- AO → A0 (Arduino UNO)

(Optional DO → digital pin for simple detection)

Optional LED

- Anode → D8 (via 2200 resistor)
- Cathode → GND

ESP8266 ^ Arduino UNO

- ESP TX → Arduino RX (Pin 0)
- ESP RX → Arduino TX (Pin 1)
- VCC → 3.3V
- GND → GND
- EN (CH_PD) → 3.3V

Power

- Arduino → USB
- ESP8266 → Stable 3.3V

Cloud Platforms

You can visualize rain data using:

- ❖ ThingSpeak
- ❖ Blynk
- ❖ Adafruit IO

Dashboard Output

- ❖ Rain intensity values
- ❖ Graph of rainfall over time
- ❖ Real-time updates

Working Explanation Step-by-Step Flow

1. Rain sensor detects water droplets
2. Arduino UNO reads analog value
3. If rain detected:
 - o LED turns ON
4. Arduino sends data via Serial
5. ESP8266 receives data
6. Connects to WiFi
7. Sends data to cloud
8. Dashboard updates rain status

System Flow

Rain Sensor → Arduino UNO → Serial → ESP82 → WiFi → Cloud → Dashboard

12.IR Remote Menu System

Components Required

- ❖ Arduino Uno
- ❖ IR Receiver Module (TSOP1838)
- ❖ IR Remote (TV/Generic)
- ❖ 16x2 LCD Display
- ❖ Jumper wires
- ❖ Breadboard

Connections IR Receiver

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ OUT → Pin 2

LCD (16x2)

- ❖ RS → Pin 7
- ❖ EN → Pin 6
- ❖ D4 → Pin 5
- ❖ D5 → Pin 4
- ❖ D6 → Pin 3
- ❖ D7 → Pin 8
- ❖ VSS → GND
- ❖ VDD → 5V
- ❖ V0 → Middle of potentiometer

Remote Buttons (Example Mapping)

Button	Function
UP	Next menu item
DOWN	Previous item
OK	Select option

Working Explanation

Step-by-Step Flow

1. User presses button on IR remote
2. IR receiver detects infrared signal
3. Arduino decodes signal value
4. Based on button:
 - o UP → Increase menu
 - o DOWN → Decrease menu
 - o OK → Select option
5. LCD updates display accordingly

System Flow

IR Remote → IR Receiver → Arduino → LCD Display

13.IR Controlled Robot Car

Components Required

- ❖ Arduino Uno
- ❖ IR Receiver Module (TSOP1838)
- ❖ TV Remote
- ❖ L298N Motor Driver
- ❖ 2 DC Motors + Wheels
- ❖ Robot Chassis
- ❖ Battery pack
- ❖ Jumper wires

Connections IR Receiver

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ OUT → Pin 2

Motor Driver (L298N)

Control Pins

- ❖ IN1 → Pin 3
- ❖ IN2 → Pin 4
- ❖ IN3 → Pin 5
- ❖ IN4 → Pin 6

Power

- ❖ 12V → Battery
- ❖ GND → Common GND

Motors

- ❖ Left Motor → OUT1 & OUT2
- ❖ Right Motor → OUT3 & OUT4

Remote Button __ Mapping (Example)

Button	Action
T (UP)	Forward
1 (DOWN)	Backward
^ (LEFT)	Left
^ (RIGHT)	Right
OK	Stop

Working Explanation

Step-by-Step Flow

1. User presses button on TV remote
2. IR remote sends infrared signal
3. IR receiver detects signal
4. Arduino decodes HEX value
5. Based on command:
 - o Motor driver controls motors
6. Robot moves in selected direction

System Flow

IR Remote → IR Receiver → Arduino → Motor Driver → Motors

14. IR Password System

Components Required

- ❖ Arduino Uno
- ❖ IR Receiver Module (TSOP1838)
- ❖ TV Remote
- ❖ Servo Motor (*or LED for demo*)
- ❖ Buzzer (optional)
- ❖ Jumper wires
- ❖ Breadboard

Connections IR Receiver

- ❖ VCC → 5V
- ❖ GND → GND
- ❖ OUT → Pin 2

Servo Motor (Lock)

- ❖ Red → 5V
- ❖ Brown/Black → GND
- ❖ Yellow → Pin 9

Buzzer (Optional)

- ❖ Positive → Pin 8
- ❖ Negative → GND

Button	Value
1	Adds "1"
2	Adds"2"
3	Adds"3"
4	Adds"4"
OK	

Example Password Entry

Press: 1 → 2 → 3 → 4 → OK

Working Explanation Step-by-Step Flow

- User presses buttons on remote
- IR receiver detects signals
- Arduino converts HEX → digits
- Stores sequence in a string
- When OK pressed:
 - Checks entered password
- If correct:
 - Servo unlocks
- If wrong:
 - Buzzer alert

System Flow

IR Remote → IR Receiver → Arduino → Password Check → Output (Servo/Buzzer)

Package Includes: 1 x Arduino IoT Starter Kit

Kit Includes:

- 1 x Arduino Uno R3
- 1 x WiFi Module ESP8266
- 1 x Bluetooth Module HC-05
- 1 x USB Cable
- 2 x 9 V Battery
- 1 x 9V Battery Clip
- 1 x Clip with DC Jack
- 1 x Breadboard
- 90 x Jumper Wires
- 1 x Resistor Kit (30 different values)
- 10 x LEDs (Red, Green, Blue, White, Yellow, RGB, 5mm)
- 3 x RGB LED
- 3 x Push Buttons
- 1 x Buzzer
- 1 x Ultrasonic Sensor HC-SR04
- 1 x Temperature & Humidity Sensor DHT11

- 1x LDR
- 1 x IR Sensor Module
- 1 x PIR Motion Sensor
- 1 x Soil Moisture Sensor
- 1 x Gas Sensor
- 1 x Sound Sensor
- 1 x Rain Drop Sensor
- 1 x Touch Sensor
- 1 x Relay Module
- 1 x Servo Motor SG90
- 1 x DC Motor
- 1 x Fan Module
- 1 x Water Pump
- 1 x Motor Driver L298N
- 2 x BO Motors (Dual Shaft, 100 RPM)
- 2 x Wheels (65 mm diameter)
- 1 x Robot Chassis (Acrylic)
- 1 x Caster Wheel
- 1 x IR Remote Control + Receiver Kit (8m range, 60° angle)
- 1 x 16x2 LCD Display with I2C (Address: 0x27)
- 1 x TM1637 4-Digit 7 Segment Display